Vijay Baskar

Principal AI Engineer | Lead GenAI & Agentic Systems Architect

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SUMMARY

Principal AI & Systems Engineer with 13+ years architecting production Generative AI, real-time computer vision, agentic pipelines, and scalable enterprise platforms. Lead GenAI and agentic systems for in-store retail: multi-agent orchestration (LangGraph, CrewAI, AutoGen, MCP, ReAct), advanced RAG (hybrid BM25, multi-query, cross-encoder rerank, Pinecone, Qdrant, Milvus, Weaviate, pgvector, FAISS), and edge vision (YOLO v8/v10/v11, RT-DETR, Kimera-VIO, OpenCV) with ONNX / TensorRT / TF Lite inference under 100ms. Shipped ezyCart self-checkout, a GenAI/RAG shopping assistant, AI recipe generation, and Lisa. Independent quantitative trading indicators for 6+ years. Supporting mobile experience in Kotlin, Java, Jetpack Compose, Flutter, Android, and iOS.

SKILLS

Generative AI & Agentic: LangGraph, CrewAI, AutoGen, LangChain, LlamaIndex, LangSmith, Langflow, MCP, Multi-Agent Orchestration, ReAct, LLM abstraction (routing, fallbacks, rate limits, token budgeting), n8n

LLMs: GPT-4o, Claude, Gemini, Llama 3 / Mistral, FinGPT, Ollama, vLLM, Prompt Engineering, evals, LoRA/PEFT, Hugging Face Transformers

RAG: Advanced RAG, Hybrid Search BM25, multi-query, cross-encoder rerank, Pinecone, Qdrant, Milvus, Weaviate, pgvector, FAISS, Supabase

CV / SLAM: YOLO v8/v10/v11, RT-DETR, Kimera-VIO, OpenCV, OCR, detection/segmentation

DL / ML: PyTorch, JAX, TensorFlow, Keras, Scikit-learn, XGBoost, NumPy, Pandas, time-series, ONNX, TensorRT, TF Lite

Backend: Python, FastAPI, Pydantic, Celery, Node.js, Spring Boot, REST, GraphQL, WebSockets, SQL, PostgreSQL, Firebase, AWS

MLOps: AWS Bedrock, SageMaker, GCP Vertex AI, Databricks, Langfuse, Docker, CI/CD for ML, Jenkins, GitHub Actions, GitLab CI

Mobile (supporting): Kotlin, Java, Jetpack Compose, Flutter, Android, iOS, Dart, MVVM

EXPERIENCE

Principal AI Engineer | Lead GenAI & Agentic Systems Architect - Retailetics SDN BHD

Kuala Lumpur, Malaysia | March 2023 - Present

Lead GenAI, agentic systems, and computer vision for Retailetics in-store stack: self-checkout, RAG assistants, and sub-100ms edge inference.

- Architected agentic and computer-vision self-checkout (ezyCart) with RT-DETR, YOLO, and Kimera-VIO; ML indoor navigation saved over $2M by replacing thousands of Bluetooth beacons.
- Shipped a GenAI/RAG shopping assistant and AI recipe generation with LangChain and LlamaIndex: cuisine suggestions, cooking video, and missing-item add-to-cart.
- Built FastAPI and Spring Boot WebSocket IoT pipelines between devices and the ezyCart Android app for real-time sync.
- Deployed ONNX and TensorFlow Lite on-device models with sub-100ms edge inference for checkout vision and on-cart assist.

Senior Staff Software Engineer - Retailetics SDN BHD

Kuala Lumpur, Malaysia | February 2021 - March 2023

Led software for ezyCart, kiosks, wayfinding, staff tools, and shopper apps.

- Spearheaded the AI-infused self-checkout smart trolley (ezyCart): architecture, stakeholder design, and a real-time socket pipeline between IoT devices and the ezyCart Android app.
- Shipped the self-checkout kiosk on the same stack (scan, bag, pay) and integrated Tap to Pay and NFC.
- Delivered ML-based indoor navigation and product finder; ran A/B and split testing across merchants.
- Built a Flutter support-staff app, Android TV cart monitor, Flutter SDK lite, ezyList, stock audit, Voice Assist, and Lisa AI shopping assistant.

Senior Software Engineer - Colan Infotech

Chennai, India | November 2012 - February 2021

Shipped 20+ Android, iOS, and Flutter apps plus the backends and ML prototypes they talk to.

- Delivered 20+ production mobile apps (Android/iOS): native Kotlin/Java and Flutter, with MVVM, Jetpack Compose, and CI/CD (Jenkins, GitLab, Bitrise); reduced ANR crashes by 70% and lifted app performance by 30%.

- Architected scalable backends using Spring Boot, AWS, Firebase, and REST APIs; designed Spring Boot microservices for 500K+ users with 40% faster response time.
- Built Python prototypes with TensorFlow, PyTorch, and Scikit-learn for on-device and backend ML experiments.

INDEPENDENT VENTURE

TAN Intelligence - Founder (not an employer)

Proprietary custom trading indicators | Independent venture

- End-to-End Quantitative Systems: Engineered and deployed proprietary algorithmic trading models and custom indicators across Commodities, FX, and Crypto, operating systematically for over 6 years.
- Automated Architecture and Telegram Pipelines: Built custom 5-minute and 15-minute signal engine pipelines integrated with Telegram alerts and low-latency automated execution AI-bots via REST and WebSockets.
- Strategy and Market Structure: Developed rule-based frameworks applying Smart Money Concepts (SMC), liquidity sweeps, and dynamic position-sizing protocols to optimize risk-adjusted returns.
- Execution and Backtesting Rigor: Automated multi-broker execution (TradingView, MetaTrader) using Python and Pine Script; backtested and forward-tested strategies to maintain a sub-5% maximum drawdown limit.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering

Anna University / GGR College of Engineering, Vellore, India | 2007 - 2011

CERTIFICATIONS

Mobile iOS & Android development - Target Software, Chennai

SOLID Principles - Coursera

AI/ML Architecture - IBM

Master prompting - Claude and Coursera